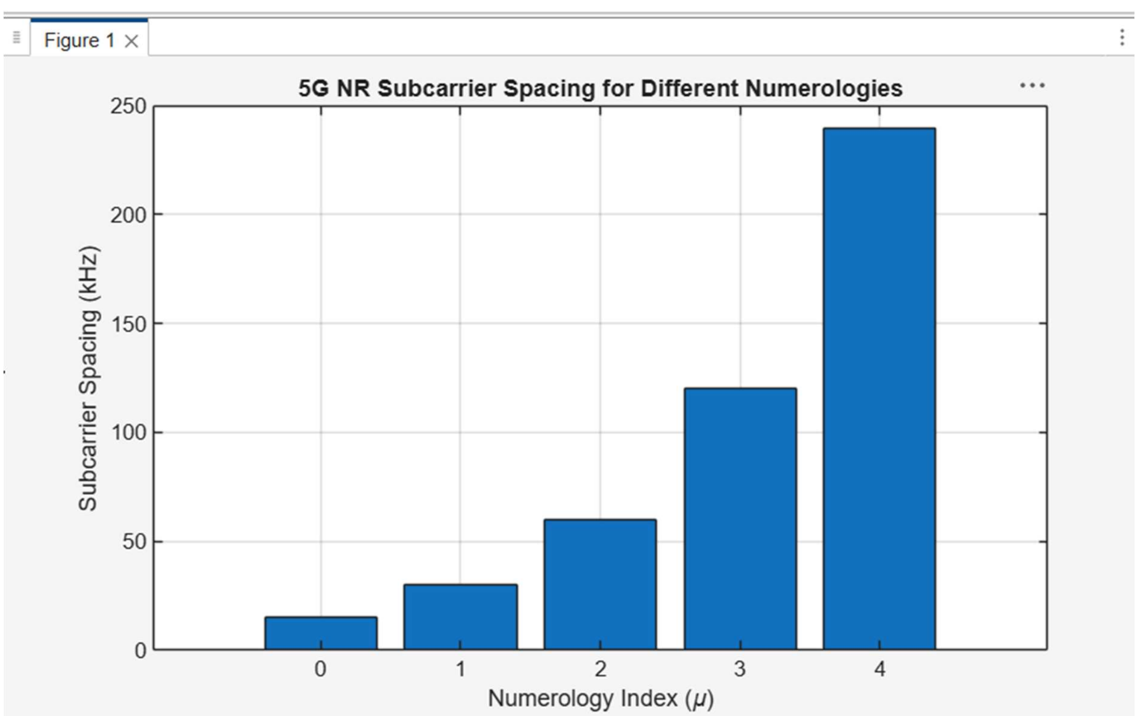
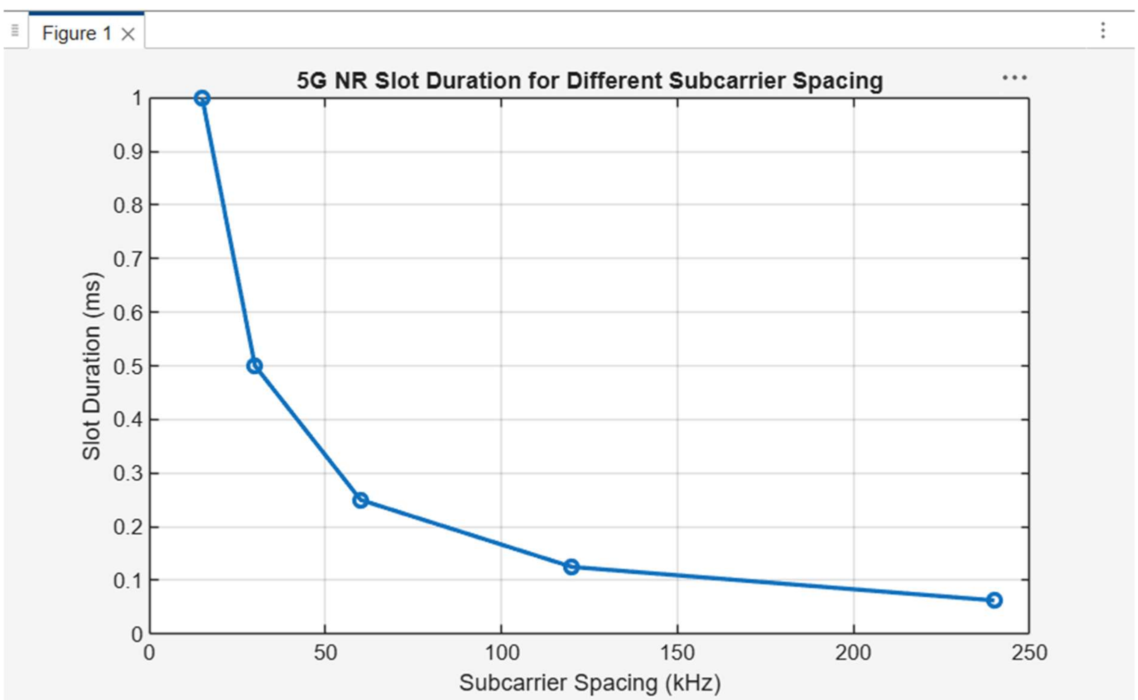
Objective 1

```
clc;

clear;

close all;

scs = [15 30 60 120 240];

mu = 0:4;

bar(scs)

grid on

set(gca,'XTick',1:5)

set(gca,'XTickLabel',mu)

% Subcarrier Spacing (kHz)

% Numerology Index

xlabel('Numerology Index (\mu)')

ylabel('Subcarrier Spacing (kHz)')

title('5G NR Subcarrier Spacing for Different Numerologies')
```



Objective 2

```
clear;

close all;

scs = [15 30 60 120 240];

% Subcarrier Spacing (kHz)

slot = [1 0.5 0.25 0.125 0.0625];  % Slot Duration (ms)

plot(scs, slot, 'o-', 'LineWidth', 2)

grid on

xlabel('Subcarrier Spacing (kHz)')

ylabel('Slot Duration (ms)')

title('5G NR Slot Duration for Different Subcarrier Spacing')
```



Objective 3

```
clc;
clear;
close all;
scs = [15 30 60 120 240];
symbols = [14 14 14 14 14];
figure
bar(scs, symbols)
% Subcarrier Spacing (kHz)
% OFDM Symbols per Slot
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Number of OFDM Symbols')
title('OFDM Symbols per Slot for Different Subcarrier Spacing')
ylim([0 16])
```